

University of Ruhuna - Faculty of Technology
Bachelor of Information & Communication Technology Honours Degree
Level 1 (Semester II) Examination, November/December 2022
Academic year 2020/2021

Course Unit: ICT 1253 Computer Networks (Written)

Duration: 2 hours

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This question paper contains five (05) pages including this instruction page.

IMPORTANT INSTRUCTIONS

- The medium of this examination is **English**.
- This is a **Closed Book** examination.
- This Examination consists of four (04) questions that are given equal marks.
- You must **answer all four (04) questions** in this examination.

1.

- a) Briefly explain the term **computer networks**.

[5 marks]

- b) Briefly explain the following two major types of computer architecture stating two (02) advantages for each type.

- i. Client server architecture
- ii. Peer-to-peer architecture

[20 marks]

- c) State the purpose of following protocols used in application layer.

- i. FTP
- ii. SMTP

[10 marks]

- d) Consider the following statement regarding the Transport layer.

"Transport layer is responsible for end-to-end transportation of data between applications".

- i. Compare **Transmission Control Protocol (TCP)** and **User Datagram Protocol (UDP)** used in transport layer considering two (02) key aspects.

[20 marks]

- ii. A three-way handshake is used to establish a TCP connection. Illustrate the main stages of the three-way handshake using a suitable diagram.

[20 marks]

e)

- i. State the Shannon's formula (Nyquist Theorem) for capacity of a noisy channel.

[10 marks]

- ii. A signal with added noise has a bandwidth of 2000 Hz assigned for data communication. The SNR is usually 2047. What will be the capacity for this channel?

[15 marks]

2.

- a) Briefly describe the purpose of an IP address.

[5 marks]

- b) Consider the two classful IP addresses given below. Identify the class of each IP address.

- i. 179.100.5.89
- ii. 15.200.255.13

[10 marks]

c) You have sub-netted your class C network 212.108.2.0 with a subnet mask of 255.255.255.224.

- i. What is the maximum number of subnets that you can create? [10 marks]
- ii. Calculate the usable hosts per subnet that can be created? [10 marks]
- iii. Identify following parameters of the first subnet
 - I. Network address (Subnet address) [5 marks]
 - II. Broadcast address [5 marks]
 - III. Usable IP range [10 marks]

d) Suppose that a company ABC Pvt (Ltd) has five (05) departments, namely, Finance, Management, Human Resources, Marketing and Sales. The network address of the company is 198.224.3.0/24.

The number of host computers in each of the department is given in Table Q2.

Table Q2

Department	No. of hosts
Finance	23
Management	5
Human Resources	48
Marketing	9
Sales	2

Using **unequal subnetting** concept, identify the following for each department

- i. Subnet mask
 - ii. Network address (Subnet address)
 - iii. Broadcast address
- [45 marks]

3.

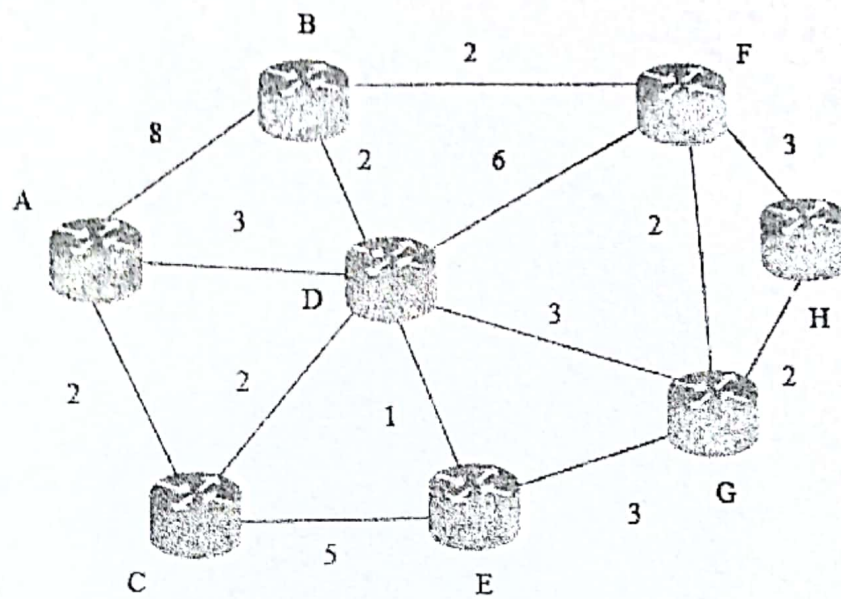
a) Consider the following statement regarding the Data Link Layer.

“Frames are the protocol data unit used in Data Link layer and framing is a function of the data link layer which provides a way for a sender to transmit a set of bits that are meaningful to the receiver”.

- i. Identify four (04) framing methods which are commonly used in Computer Networks.

[8 marks]

- ✓ ii. Suppose that the number of bits allocated for header, payload and trailer of a fixed size frame is 12, 25 and 12, respectively. Calculate the required number of frames, if 275 bits of actual data are to be transmitted as frames.
[10 marks]
- b) Briefly explain the statement "Data link layer is responsible for error control".
[5 marks]
- c) Suppose that the received sum of data at the receiver's side is 11010101 when the transmitted data unit by the sender is 10011001 10100010 00110100 10000001. Assuming that an **8-bit checksum** is used,
- i. Calculate the checksum value generated at the sender's side.
[20 marks]
- ii. Justify the existence or non-existence of an error.
[10 marks]
- d) Suppose that a received code word is 1111011 when **7-bit hamming code** is used. Assuming the even parity, identify whether there is an error with the received code word and locate the error if there is any.
[35 marks]
- e) Identify difference between **unacknowledged connectionless service** and **acknowledged connected oriented service** considering three (03) key aspects.
[12 marks]
- 4.
- a) Consider the following statement regarding the Network Layer
"The network layer is responsible for routing packets".
- i. Briefly explain the purpose of routing algorithms.
[5 marks]
- ii. Identify and describe three (03) common requirements that need to be fulfilled by routing algorithms.
[15 marks]
- iii. Identify the differences between **static routing algorithms** and **dynamic routing algorithms** considering two (2) key aspects.
[20 marks]
- b) Consider the following network topology in which the cost of each link is given. Find the **shortest path** from **Router A** to every other router in the topology by using **link state routing algorithm**. Provide your steps using a table.



[60 marks]

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